HW 7.

1.

Given $q(x,y)$ is irreducible and symmetric, there exists a path from any state x to any states y with non-zero probability under q. The Metropolis Algorithm states that:

$$p(x,y) = \begin{cases} q(x,y) \cdot \min\left\{1, \frac{\pi(y)}{\pi(x)}\right\}, & \text{if } x \neq y \\ 1 - \sum_{z \neq x} p(x,z), & \text{if } x = y \end{cases}$$

$\pi(x) > 0$ and $\pi(y) > 0$ for all $x, y \in \mathcal{X}$ (by the assumption) ①

$q(x,y) > 0$ because of q is irreducible. ②

$\min\left\{1, \frac{\pi(y)}{\pi(x)}\right\}$ will always be positive because of ①.

When $x \neq y$, $p(x,y) = \underbrace{q(x,y)}_{>0} \cdot \underbrace{\min\left\{1, \frac{\pi(y)}{\pi(x)}\right\}}_{>0}$

so $p(x,y) > 0 \Rightarrow p$ is irreducible.

Now consider $x = y$.

$$p(x,x) = 1 - \sum_{z \neq x} p(x,z)$$

$\sum_{z \neq x} p(x,z)$ represent the total probability of transitioning from state x to all other state $z \neq x$.

HW 7

1. $P(x, z) > 0$ for all $x \neq z$ and p is a probability function, $\sum_{z \neq x} P(x, z)$ must be bounded between 0 and 1. It can't really be 1 since if it does, it would imply that the probability of leaving state x for any other state is 100%, leaving no probability mass for staying in state x. Therefore, $\sum_{z \neq x} P(x, z)$ must be strictly less than 1. Consequently,

$P(x, x) = 1 - \sum_{z \neq x} P(x, z)$ is strictly larger than 0.

Thus, when $x = y$, $P(x, y) = P(x, x) > 0 \Rightarrow p$ is irreducible.

Now consider aperiodicity.

Since we have proved irreducibility, $d_1 = d_2 = \cdots d_s = d$ for all irreducible HMC. A MC is aperiodic is there is no fixed cycle that the chain is constrained to follow. More formally, a state x is a MC has period d if any return to state x must occur in multiples of d time steps. If $d = 1$ for every state in the MC, the MC is aperiodic, meaning that there is no fixed number of steps after which returns to a particular state must occur.

To prove the MC defined by $p(x, y)$ is aperiodic, we need to show that there exist at least one state x for which period $d = 1$. From previous proof of irreducibility when $x = y$, we have

HW7

1.

Shown the $p(x, x) > 0$, indicating that the chain has a positive probability of staying in the same state in one step. If the set of times at which a chain can return to state x is T_x, then the period d of state x is the greatest common divisor (e.c.d). $p(x, x) > 0 \Rightarrow 1$ is a member of T_x, it further indicates the GCD of the set T_x must be 1. Therefore, for state x, the period $d = 1$. Thus, $p(x, y)$ is aperiodic.

2. Proof claim 1.3.

For any $x, y \in \mathcal{X}$.

if $x = y$, $\pi(x) p(x, y) = \pi(y) p(y, x)$ is true for sure.

if $x \neq y$, we plug in $p(x, y) = q_p(x, y) \cdot \min\left\{1, \frac{\pi(y)}{\pi(x)}\right\}$, ~~the left side of equation~~

$\pi(x) p(x, y) = \pi(x) q_p(x, y) \cdot \min\left\{1, \frac{\pi(y)}{\pi(x)}\right\}$

left side of the equation $= \pi q_p(x, y) \cdot \min\{\pi(x), \pi(y)\}$ ①

$\pi(y) p(y, x) = \pi(y) q_p(y, x) \min\left\{1, \frac{\pi(x)}{\pi(y)}\right\}$

$= q_p(y, x) \min\{\pi(y), \pi(x)\}$ ②

Because $q_p(x, y) = q_p(y, x)$ for all $x, y \in \mathcal{X}$. ① = ②

Successfully proved claim 1.3.